Abdul Moeez

Date of birth: 24/12/2004 | Place of birth: Lahore, Pakistan | Gender: Male | Nationality: Pakistani | Phone: (+92) 3090358869 (Mobile) | Email address: moeezch095@gmail.com | Website: https://github.com/moeezch095?tab=projects | Address: 48 GT road near shazoo lab 119D, 48 GT road near shazoo lab 119D, 51000, lahore, Pakistan (Home)

About myself

Backend-focused Computer Science student with hands-on experience in NestJS, Node.js, and MongoDB, currently working on an IoT-based energy monitoring system. Skilled in REST API development, testing, role-based access control, and real-time data dashboards.

Alongside backend development, I have a solid foundation in front-end technologies including HTML, CSS, JavaScript, and React.js. Known for strong communication skills, quick learning ability, and the capability to collaborate effectively while building clean, user-focused applications.

Education & Training

Bachelor of Computer Science | The Superior University | Lahore, Pakistan

Intermediate | Govt Islamia Graduate College Civil Lines, Lahore. | Lahore, Pakistan

Work experience

Web Application Developer (MERN) | Lahore, Pakistan

- Website Planning & Design
- Front-End Development
- Back-End Development
- E-Commerce Features

Implement product filters, sorting, and inventory management.

Backend Developer | Lahore, Pakistan

- Developed RESTful APIs using **Node.js** and **NestJS**.
- Built and connected server-side logic with MongoDB databases.
- Designed efficient backend structures including controllers, services, and schemas.
- Tested APIs using **Postman** for request validation, debugging, and performance checks.
- Collaborated on full-stack projects using **React.js** for front-end integration.
- Worked with Git and GitHub for version control and teamwork.

Backend Developer (IOT Systems) | 2026 - Current | Lahore, Pakistan

• Worked as a Backend Developer on an IoT-based Energy Monitoring Platform using **NestJS** and **MongoDB**.

• Designed and developed multiple backend modules including **Dashboards, Trends, Production, Reports, and User Management**.

• Implemented real-time and historical data handling for live energy consumption and production monitoring.

• Built and optimized MongoDB aggregation pipelines to process high-volume IoT sensor data for dashboards and analytics.

• Developed RESTful APIs for live data visualization, historical trends, and reporting features.

• Implemented secure authentication and authorization using JWT-based Role-Based Access Control (RBAC).

• Designed and managed User Management module including roles, privileges, and access control policies.

• Engineered real-time data workflows to support live charts and energy consumption tracking.

• Ensured scalable backend architecture for handling continuous IoT data streams.

• Used Postman for API validation, debugging, and performance testing.

Projects

Doctor–Patient Appointment Management System

Technologies: Node.js, Express.js, MongoDB, Mongoose, JWT Authentication, Middleware, Postman

Developed a backend system that allows doctors and patients to manage appointments efficiently. Implemented secure authentication using **JWT**, added custom **middleware** for role-based access, and structured modular APIs for appointments, users, and authentication. Integrated MongoDB for data storage with well-designed schemas and proper validation.

Key Features & Responsibilities:

- Designed and developed RESTful APIs using **Node.js** and **Express.js**
- Implemented **JWT-based authentication** (login, register, protected routes)
- Created custom **middleware** for authorization and request validation
- Managed database operations using **MongoDB + Mongoose**
- Built CRUD endpoints for appointments, doctors, and patient profiles
- Performed full API testing & debugging using **Postman**
- Ensured clean folder structure (controllers, routes, models, middleware, config)

IOT-Based Energy Monitoring & Analytics System

Developed a complete backend system for an IoT-based energy monitoring platform using NestJS and MongoDB. The system processes real-time sensor data and provides analytical insights through dashboards and reports.

Key Modules Developed:

• **Dashboards Module** – Implemented APIs to display live energy consumption and production data using real-time charts and metrics.

• **Trends Module** – Built historical data analytics using MongoDB aggregation pipelines for energy insights and reporting.

• **Production Module** – Developed real-time and historical production monitoring with data filtering and time-range queries.

• **User Management Module** – Designed and implemented role-based access control (RBAC), user roles, and privilege management using JWT authentication.

• **Reports Module** – Created APIs to generate structured energy reports based on time intervals and usage patterns.

• Implemented optimized aggregation pipelines for efficient processing of large IoT datasets.

• Designed scalable backend architecture capable of handling continuous real-time data streams.

Technologies Used:

- NestJS | MongoDB | JWT | REST APIs | Aggregation Pipelines | Postman

Domino Game – Backend API Testing

Technologies:

Core PHP, REST APIs, Postman

- Performed backend API testing for an online Domino game developed in Core PHP.
- Tested game-related APIs for functionality, data validation, and response accuracy.
- Identified and reported API issues related to game flow and backend logic.
- Ensured stability and correctness of backend endpoints through manual API testing using Postman.

Cars Dealing and accessories

Technologies: HTML, CSS, JavaScript, PHP Laravel

• Developed a full-featured car trading platform where users can buy and sell cars or vehicles.

• Integrated real-time chat functionality for user communication.

• Designed and implemented an e-commerce store for car gadgets and accessories.

• Used PHP Laravel to handle backend operations, including user authentication and data management.

Threads Hub

Technologies: HTML, CSS, JavaScript, PHP

• Created an interactive online store for hoodies, allowing customers to explore collections, read reviews, and make purchases with ease.

• Designed and developed responsive UI components to ensure effortless navigation and secure checkout processes for our hoodie collection, providing customers with a smooth and enjoyable shopping experience

• Utilized PHP for backend logic and data handling to efficiently manage product inventory, orders, and customer data for our hoodies clothing website.

Skills

C++ | AI Using | customer service | Microsoft Excel | Social Media | CSS | Javascripts | Begginer JavaScript for WebDevelopment | Python Language - Basic knowledge | Experience with PHP, SQL, HTML, CSS, JavaScript | Html, Java, CSS | Python (computer programming) | JavaScript | Microsoft Office | Microsoft Word | Node JS | Nest JS | NEXT JS | Javascript (React.js) | API Manual Testing and Automation Testing with Postman and Newman | Deployment with Netlify, Vercel | MongoDb, Firebase | IoT Backend Systems | Role-Based Access Control (RBAC)

Soft Skills

Soft Skills

1. Communication
2. Time management
3. Problem Solving

Language Skills

Mother tongue(s): **Urdu**

	Understanding		Speaking		Writing
	Listening	Reading	Spoken production	Spoken interaction	
English	B1	B2	A2	B1	B1